

A positive lower metric bound passes to the derivative

MathlibAnnex.BilipschitzOrientation.fderiv_lower_bound

theorem

Preserves the same lower constant in the difference-quotient limit.

Statement

Let $n \in \mathbb{N}_0$ and let $E = \mathbb{R}^n$ have the sup norm $\|\cdot\|_\infty$. Let $S, T \subseteq E$ be open, and let $f, g : E \rightarrow E$ be globally Lipschitz maps whose restrictions $f|_S : S \rightarrow T$ and $g|_T : T \rightarrow S$ are mutually inverse. Fix a number $\ell > 0$ such that

$$\ell \|x - x'\|_\infty \leq \|f(x) - f(x')\|_\infty \quad (x, x' \in E).$$

Suppose f is differentiable at a point $x \in E$. Let $Df(x) : E \rightarrow E$ denote its Fréchet derivative there. Then

$$\ell \|v\|_\infty \leq \|Df(x)v\|_\infty \quad (v \in E).$$

Assumptions

The map f is differentiable at this particular x ; the point x need not lie in S . The sets $S, T \subseteq E$ are open. The inverse and image conditions are

$$\begin{aligned} f(S) &\subseteq T, & g(T) &\subseteq S, \\ g(f(x)) &= x \quad (x \in S), & f(g(y)) &= y \quad (y \in T). \end{aligned}$$

The global bounds use specified constants $C_f, C_g \geq 0$ and $\ell > 0$:

$$\ell \|x - x'\|_\infty \leq \|f(x) - f(x')\|_\infty \leq C_f \|x - x'\|_\infty \quad (x, x' \in E),$$

$$\|g(y) - g(y')\|_\infty \leq C_g \|y - y'\|_\infty \quad (y, y' \in E).$$

The estimates hold on all of E , not just on S or T . The inverse identities are required only on S and T .

Conclusion

The derivative has the same positive lower constant ℓ as the original map. In particular,

$$Df(x)v = 0 \implies \ell \|v\|_\infty \leq 0 \implies v = 0.$$

Thus $Df(x)$ is injective. The estimate also includes the direction $v = 0$.

Applying the same estimate to $u - v$ proves that $Df(x)u = Df(x)v$ forces $u = v$, as in the derivative-injectivity corollary. Since $Df(x)$ is a linear endomorphism of the finite-dimensional space E , the determinant corollary gives $\det Df(x) \neq 0$. Similarly, the original lower bound gives global injectivity of f by applying it when $f(x) = f(x')$.

Proof route

Apply the lower bound to $x + tv$ and x , divide by $|t|$, and pass to the derivative as $t \rightarrow 0$.

▼ Proof steps

1. **Write the difference-quotient estimate.** Fix $v \in E$. For every real $t \neq 0$,

$$\begin{aligned} \ell |t| \|v\|_\infty &\leq \|f(x + tv) - f(x)\|_\infty, \\ \ell \|v\|_\infty &\leq \left\| \frac{f(x + tv) - f(x)}{t} \right\|_\infty. \end{aligned}$$

The first line is the global lower bound at these two points. The second follows from $|t| > 0$ and homogeneity of the norm; it does not require $v \neq 0$.

2. **Take the limit.** Differentiability at x gives

$$\frac{f(x + tv) - f(x)}{t} \longrightarrow Df(x)v \quad (t \rightarrow 0, t \neq 0).$$

Consequently,

$$\ell \|v\|_\infty \leq \lim_{t \rightarrow 0, t \neq 0} \left\| \frac{f(x + tv) - f(x)}{t} \right\|_\infty = \|Df(x)v\|_\infty.$$

This is precisely the difference-quotient lemma with linear map $Df(x)$, lower constant ℓ , the displayed global estimate, and differentiability at x . No derivative at any other point is needed.

Main citations

- Exact declaration and its source — `MathlibAnnex.BilipschitzOrientation.fderiv_lower_bound`
- Difference quotients and their derivative limit —
`MathlibAnnex.BilipschitzOrientation.fderiv_lower_bound_of_global_lower`
- The lower bound forces global injectivity —
`MathlibAnnex.BilipschitzOrientation.BilipschitzOpenData.f_injective`
- A positive derivative bound gives injectivity —
`MathlibAnnex.BilipschitzOrientation.fderiv_injective`
- An injective square derivative has nonzero determinant —
`MathlibAnnex.BilipschitzOrientation.det_fderiv_ne_zero`

Lean source signature (exact)

```
theorem fderiv_lower_bound {n : ℕ} (D : BilipschitzOpenData n)
  {x : Fin n → ℝ} (hdf : DifferentiableAt ℝ D.f x) (v : Fin n → ℝ) :
  D.lower * ‖v‖ ≤ ‖(fderiv ℝ D.f x) v‖
```

In the source	Mathematical meaning
<code>(D : BiLipschitzOpenData n)</code>	The record described in this Card: open $S, T \subseteq E = \mathbb{R}^n$, maps $f, g : E \rightarrow E$ restricting to inverse bijections $S \leftrightarrow T$, global upper Lipschitz bounds C_f, C_g , and global lower bound $\ell \ x - x'\ _\infty \leq \ f(x) - f(x')\ _\infty$ with $\ell > 0$. The source name <code>D</code> is not the derivative symbol.
<code>D.source; D.target; D.f; D.g</code>	The local mathematical objects S, T, f, g , respectively, throughout this Card.
<code>{x : Fin n → ℝ} (hdf : DifferentiableAt ℝ D.f x)</code>	The input point $x \in E$ is a differentiability point of f . There is no condition $x \in S$.

In the source	Mathematical meaning
<code>(v : Fin n → ℝ)</code>	The arbitrary vector $v \in E$.
<code>D.lower * v ≤ (fderiv ℝ D.f x) v </code>	The output $\ell \ v\ _\infty \leq \ Df(x)v\ _\infty$. The derivative of the same f keeps its global positive lower constant.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 23–29:

```

noncomputable section

open Set MeasureTheory Filter
open scoped ENNReal NNReal Topology

namespace MathlibAnnex
namespace BilipschitzOrientation

```

▼ Exact content identity

Declaration: `MathlibAnnex.BilipschitzOrientation.fderiv_lower_bound`
Accepted content SHA-256: `d13cfea772f82cdc47f7ef8bc5877579187689451484645ae6d6131d5c8d8420`
Accepted source guide SHA-256: `ceb153037636e289a4e391d81836cd09a8b48fac700bd2a2c1c8c0d9bda9eeb4`
Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`